

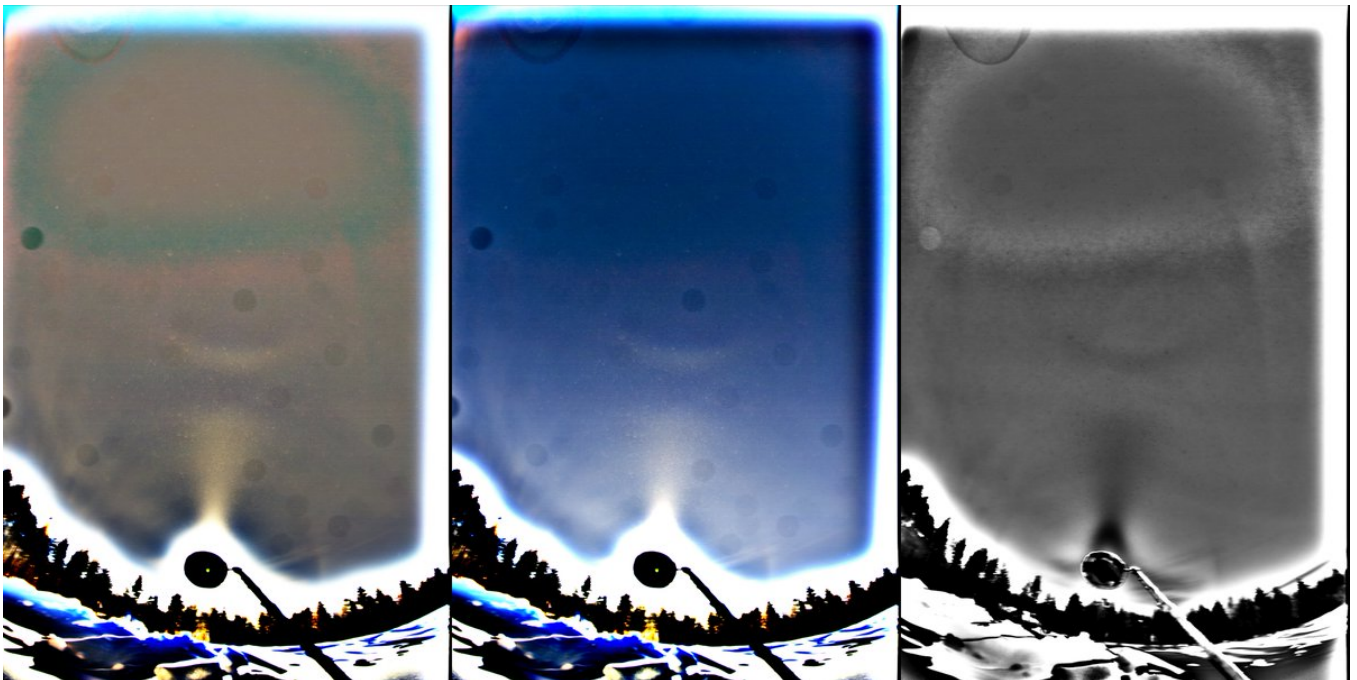
Thread - 2024-02-22

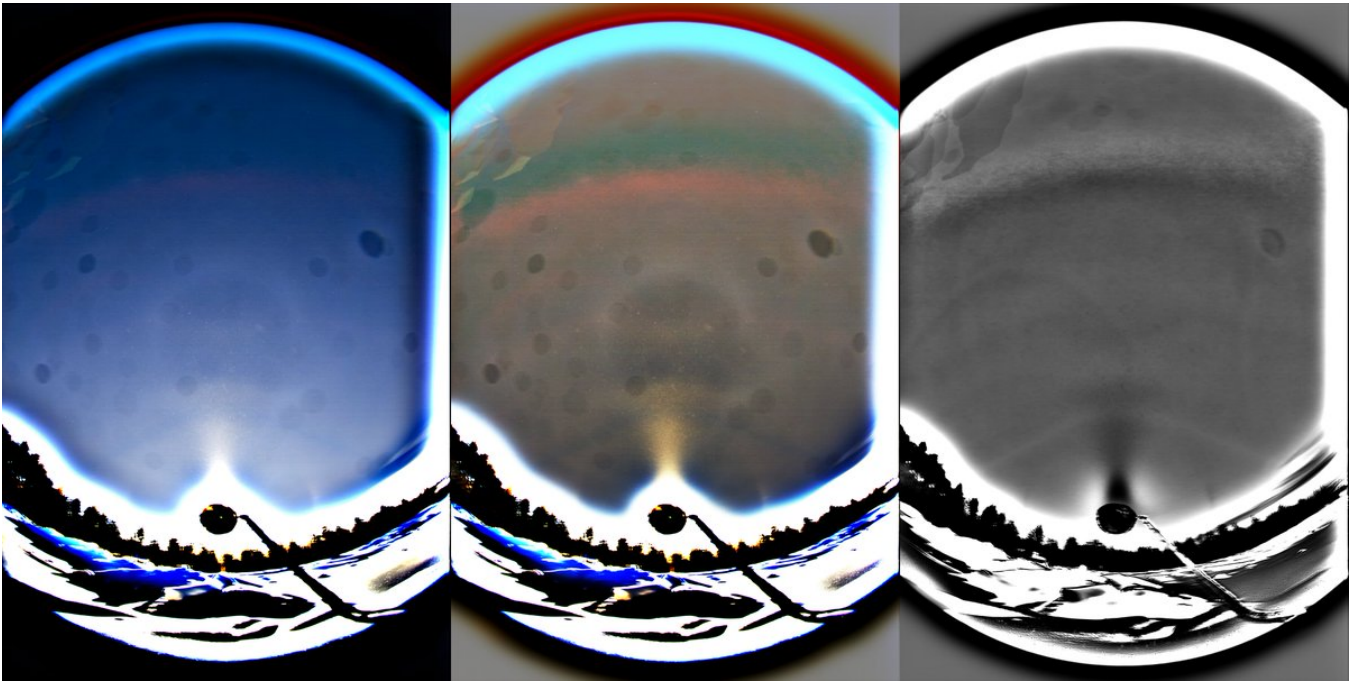
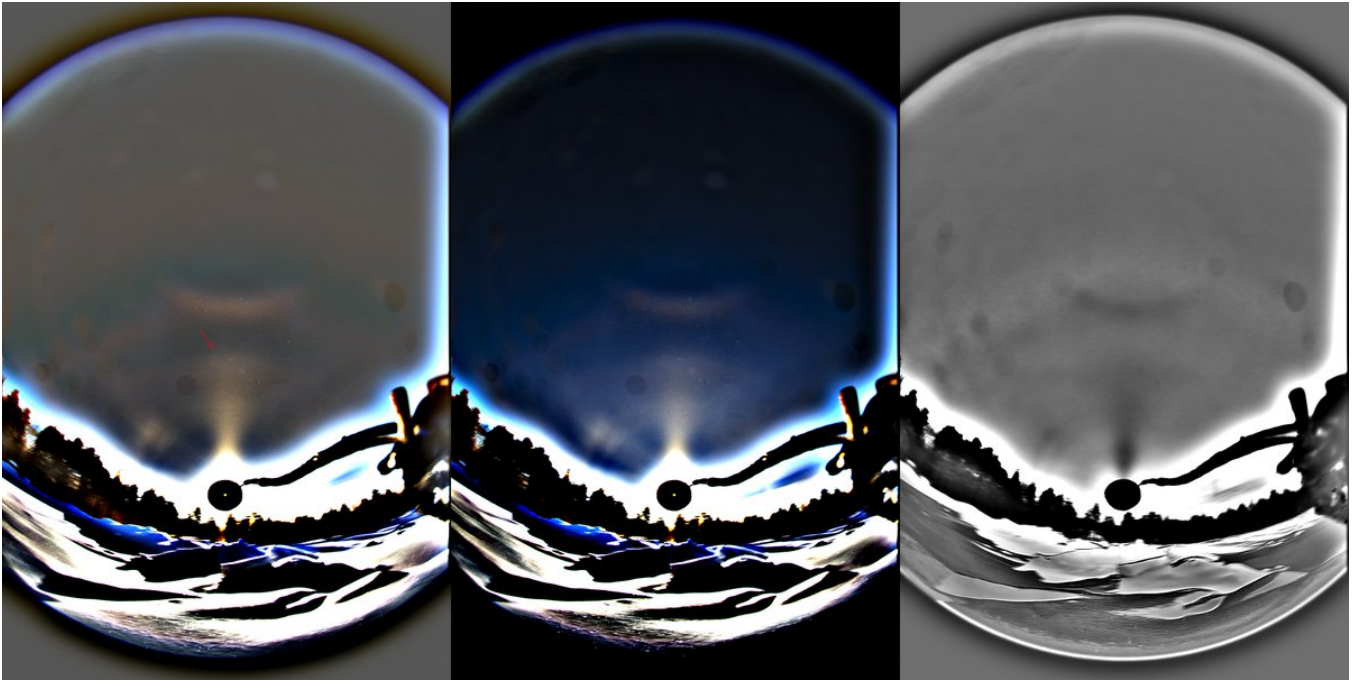
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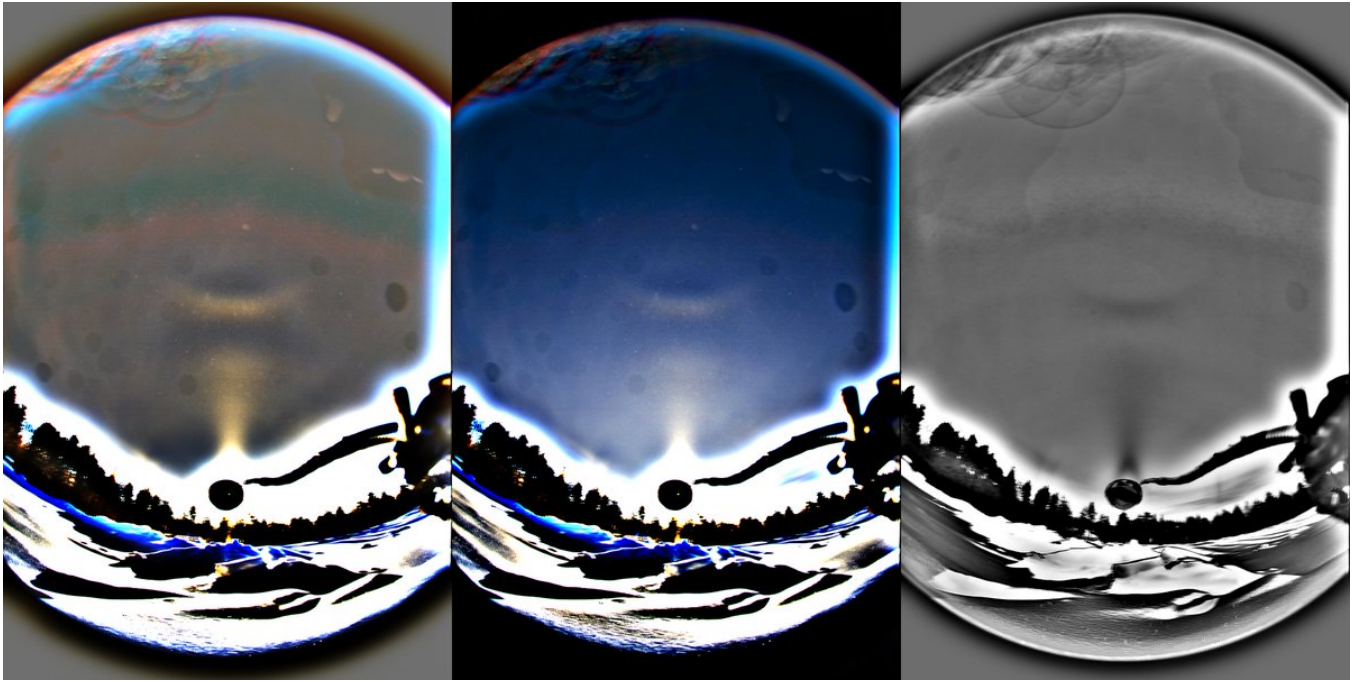
1/6 - 2024-02-22 16:19 UTC

I wasn't entirely imagining the thing in the solar vertical in the 19 February Suosiola plume display (paragraph 8 in the diary note: <https://t.co/Zyk5XXoSfW>). Stacks indeed show that the pillar ends abruptly somewhere around 35 degrees above the sun (red arrow in the first image). The four images here are stacks – in chronological order – of 151, 105, 158 and 122 frames, 2s interval (the stuff was blowing by quite fast). In b-r + bgr versions the pillar is continuous but colored photos (bgr and usm) give an impression that there is a gap in the pillar rendering the top part a separate chunk. Is that real or treatment artefact (double usm with around 105 radius, amount 500% and 300%; bgr gaussian blur radius also 105) I am not sure.

Once I get to it, I will thread more material here, such as some single photos for reference. It needs to be also determined how much the parhelia that I on the spot mainly chalked up to the background high cloud were contributed by the plume. The stacks actually show the former gave an odd radius display, I will make a bespoke stack.





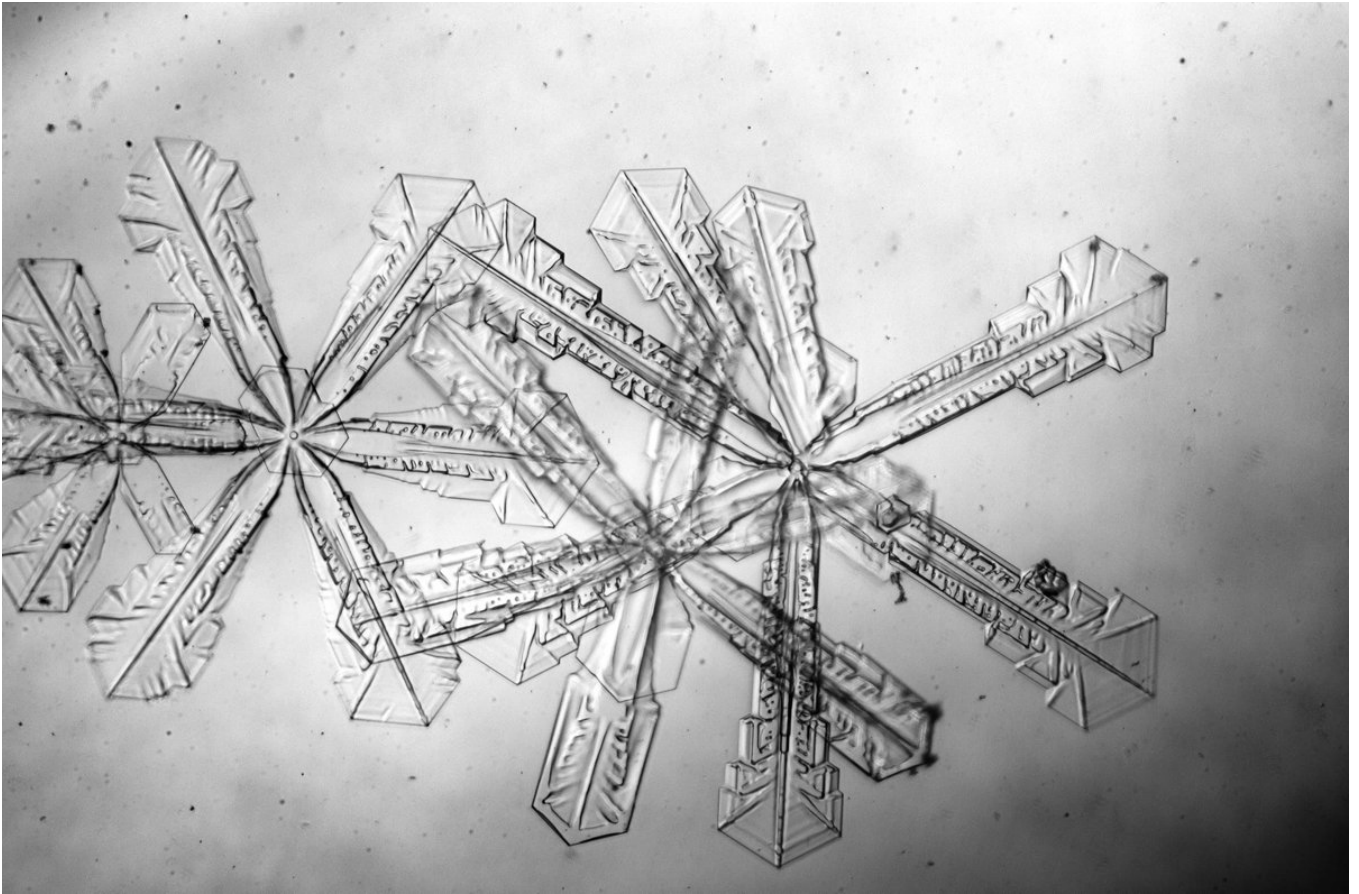


2/6 - 2024-02-23 10:34 UTC

In a meanwhile, two max stacks and crystals of the 19 February Suosiola plume display.

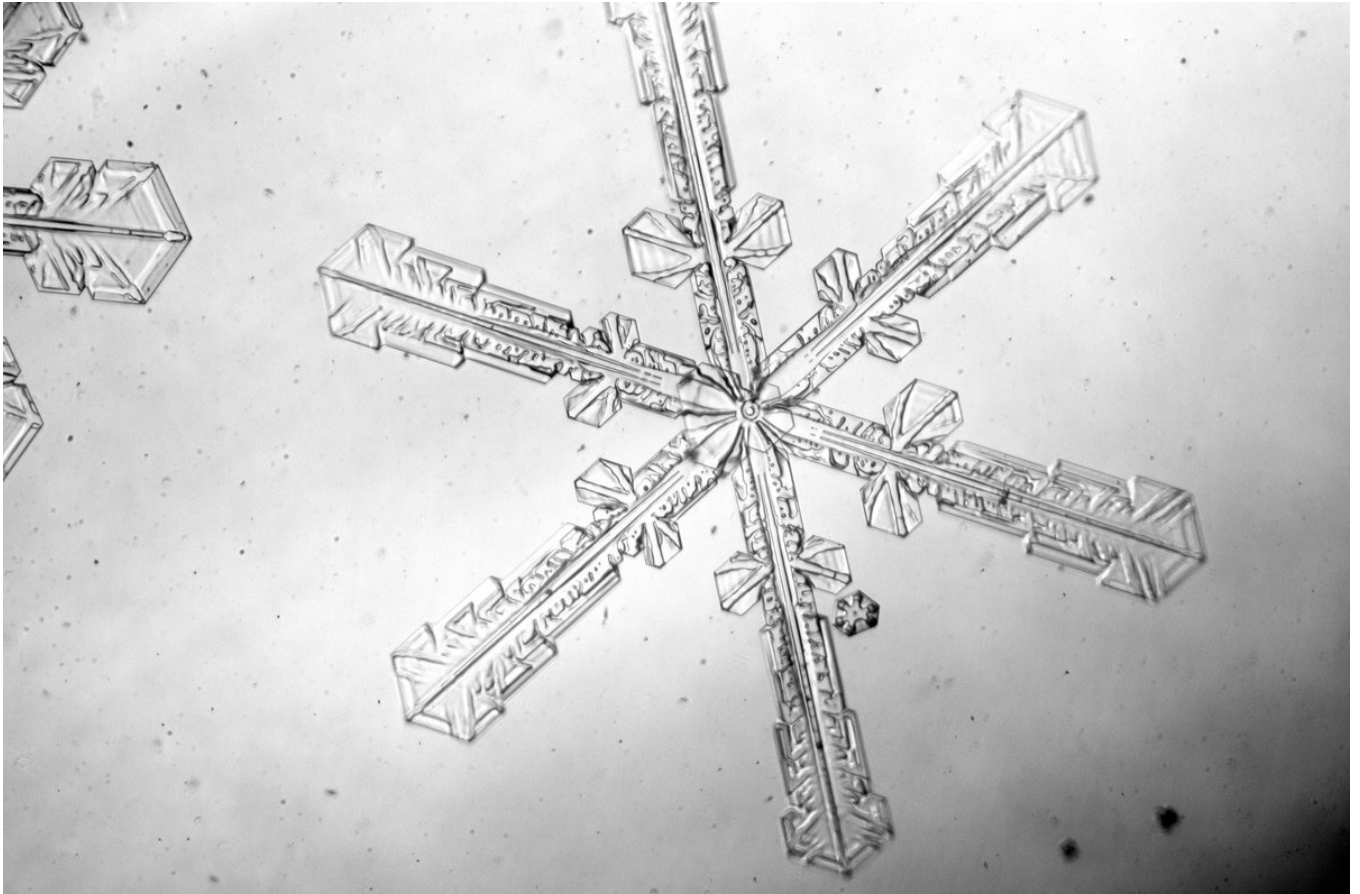


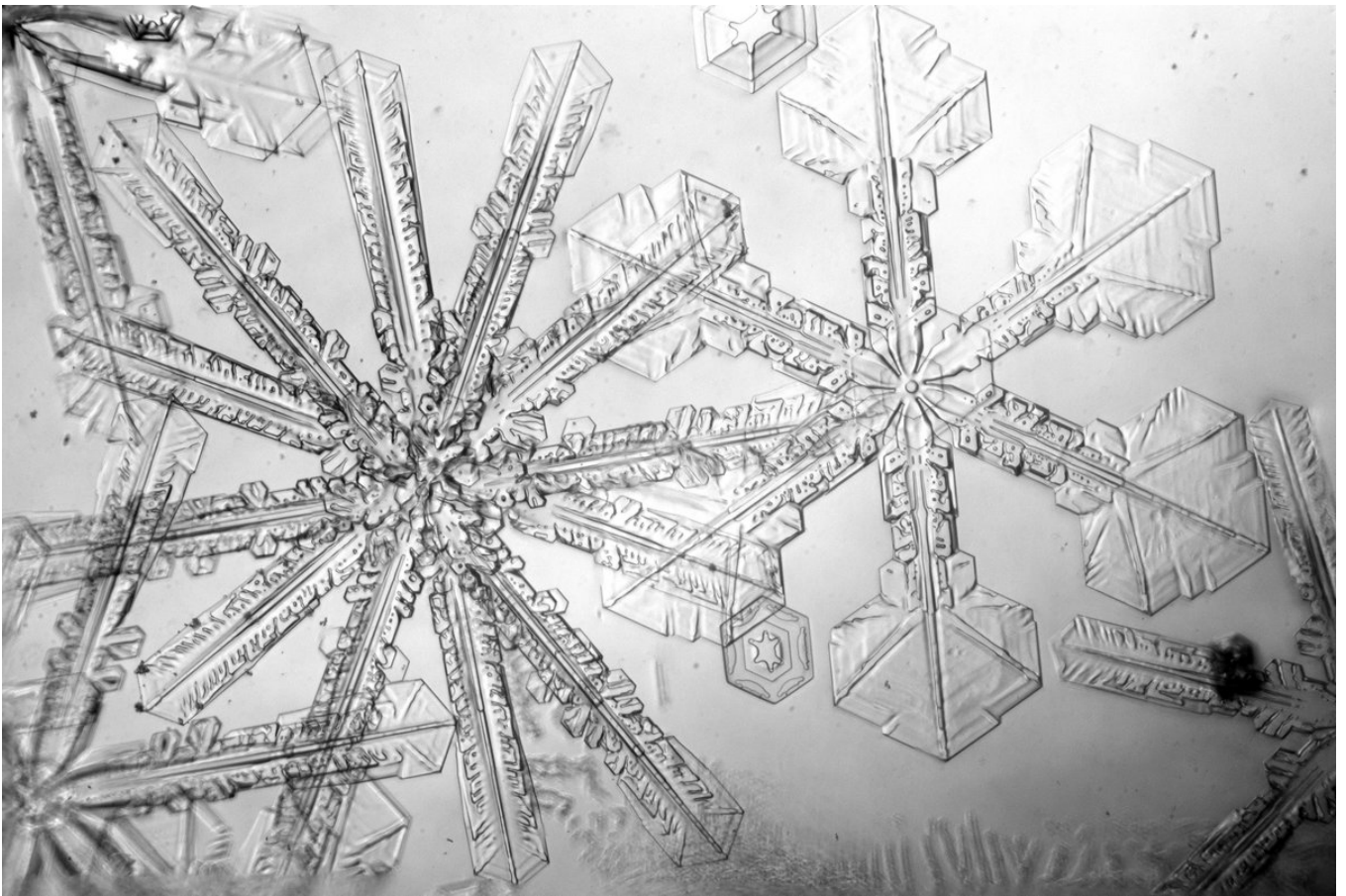
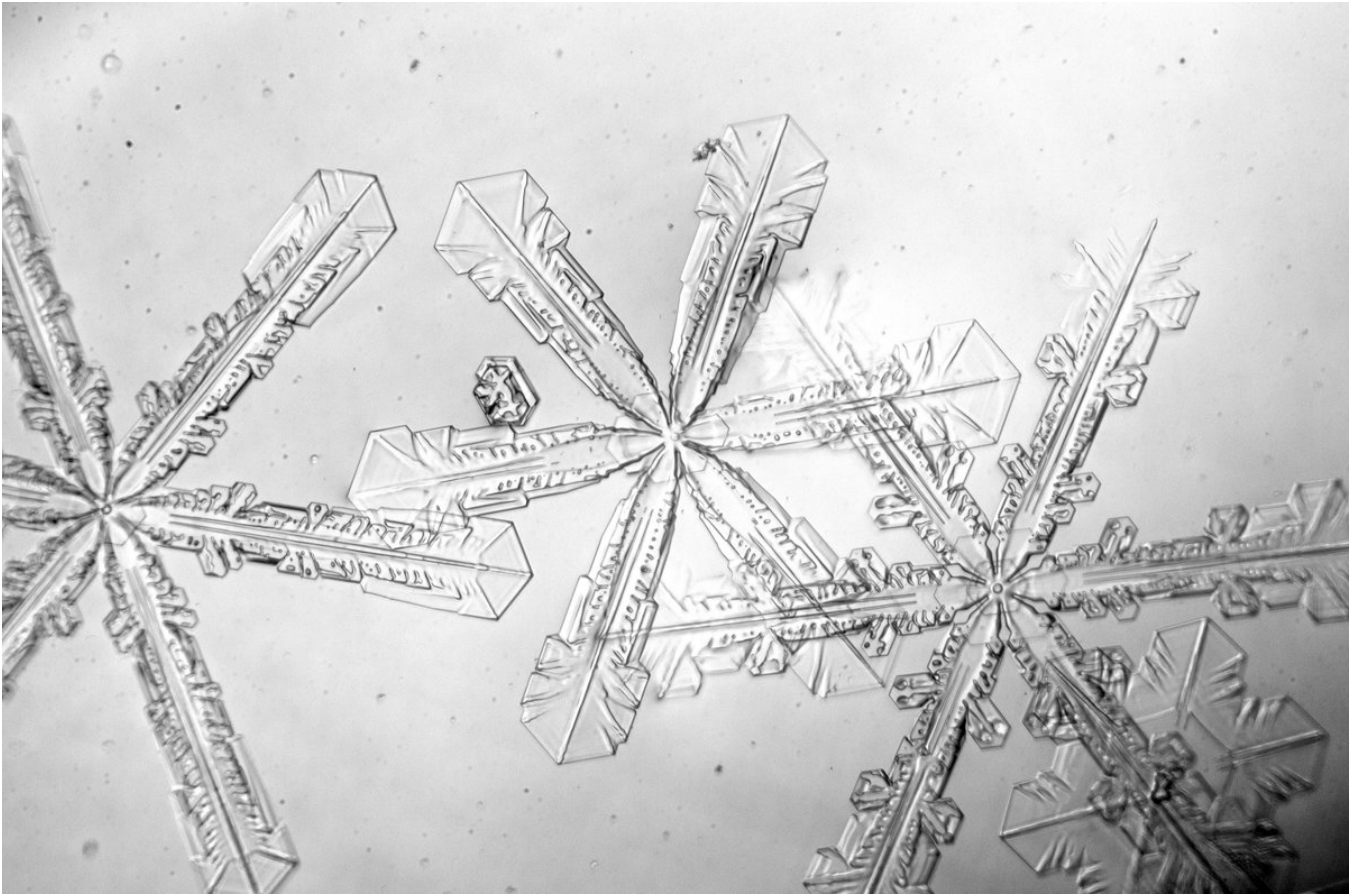




3/6 - 2024-02-23 10:36 UTC

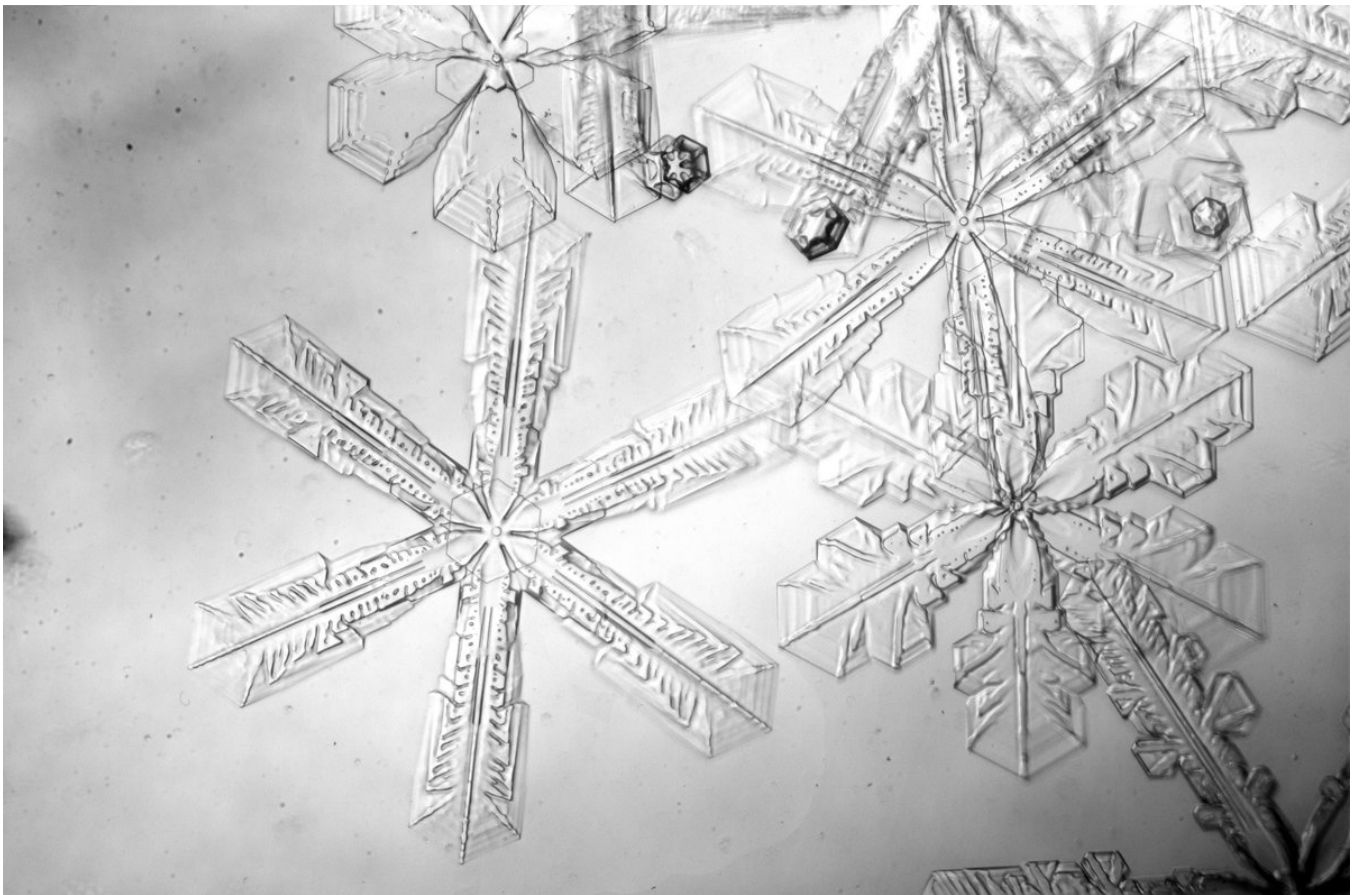
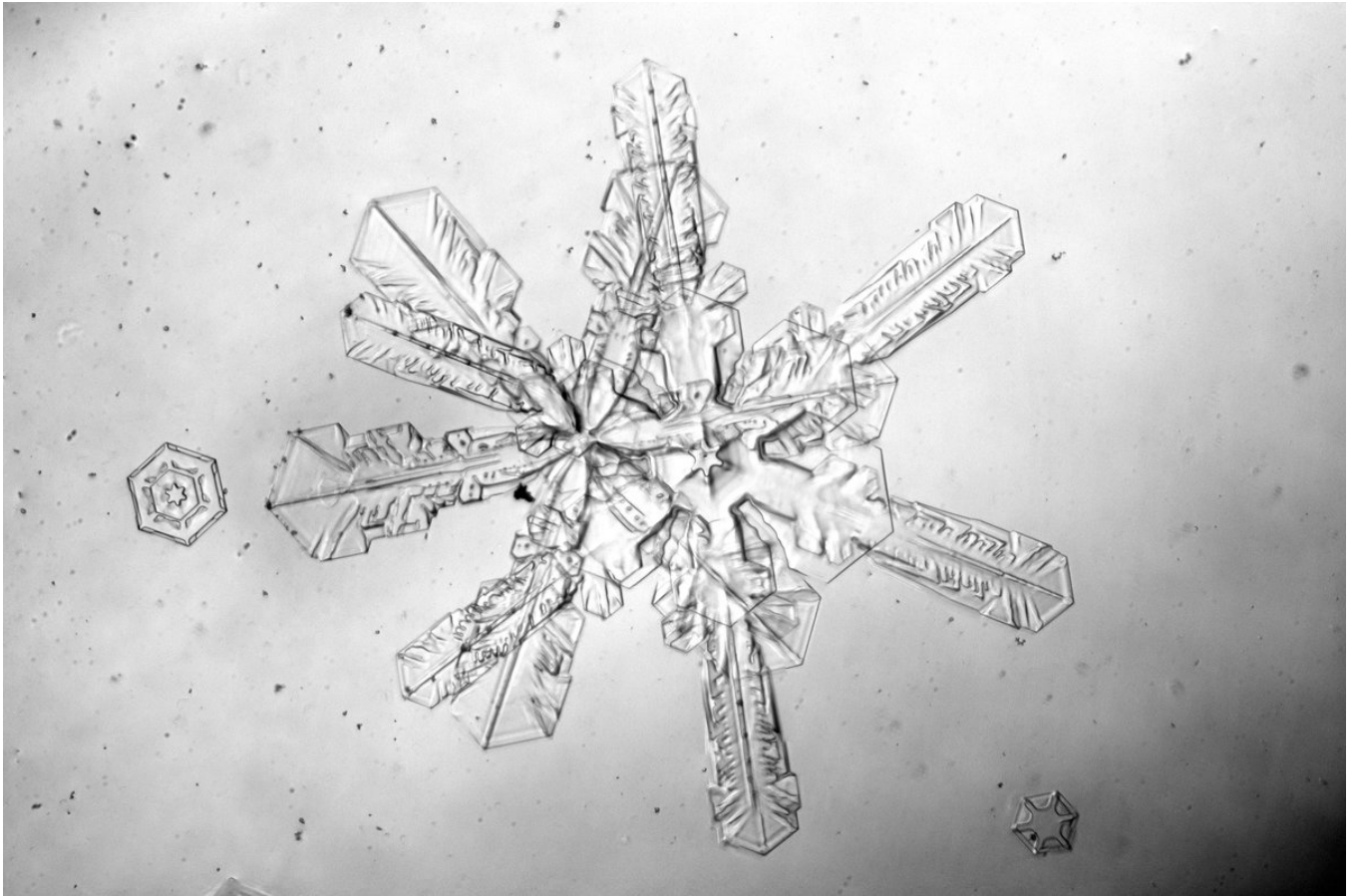
More crystals, 19 February Suosiola plume display.

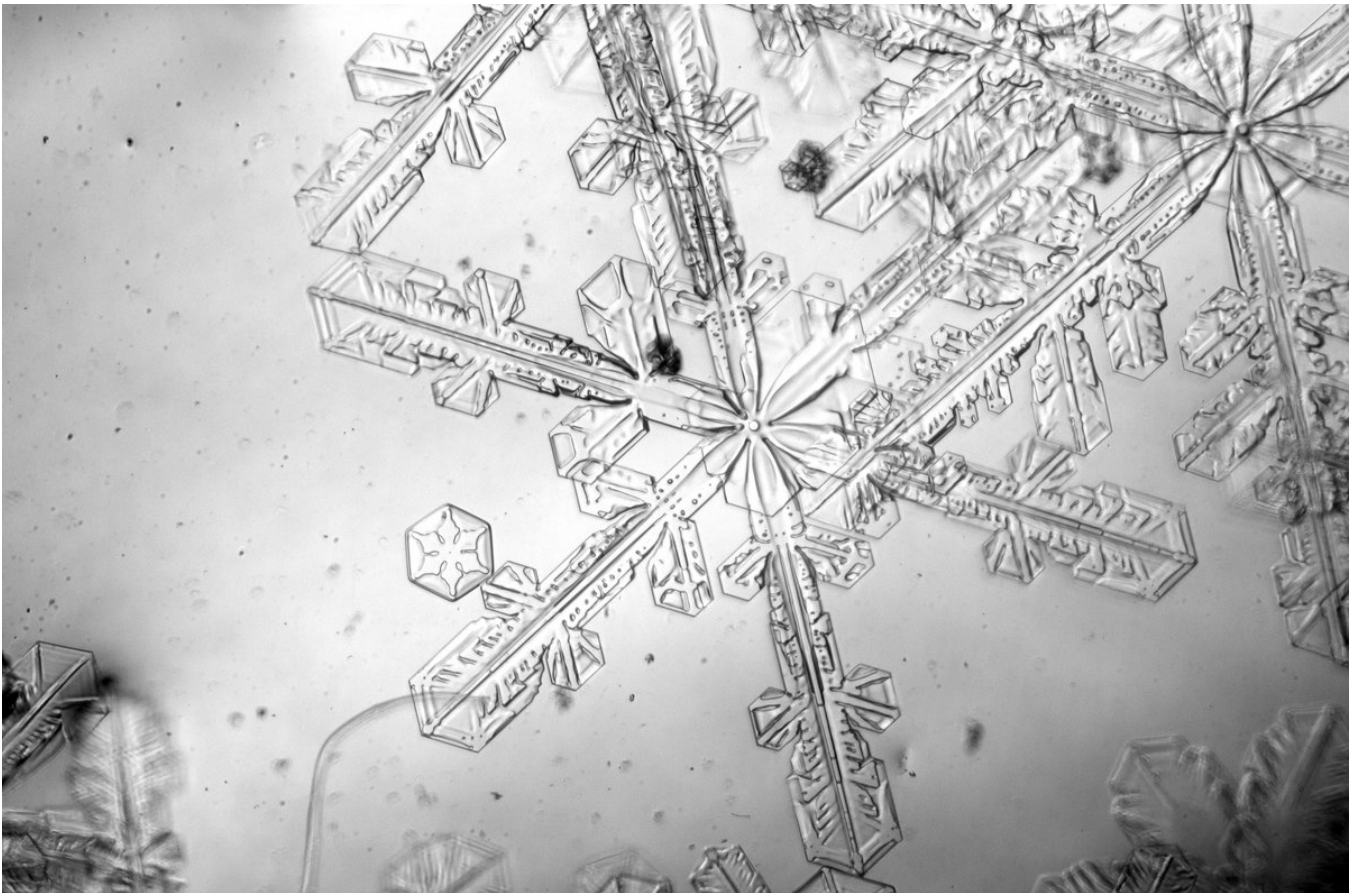
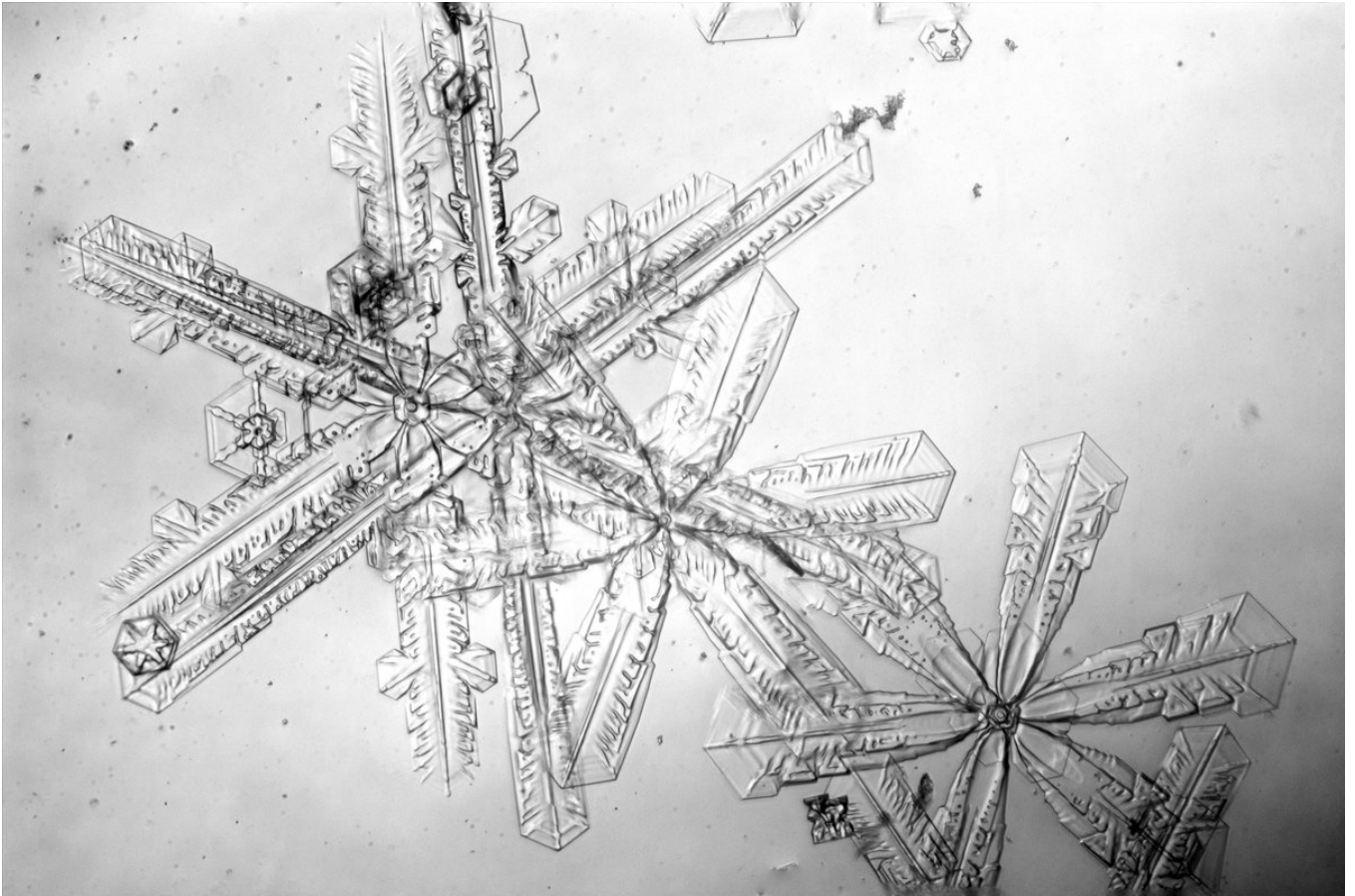




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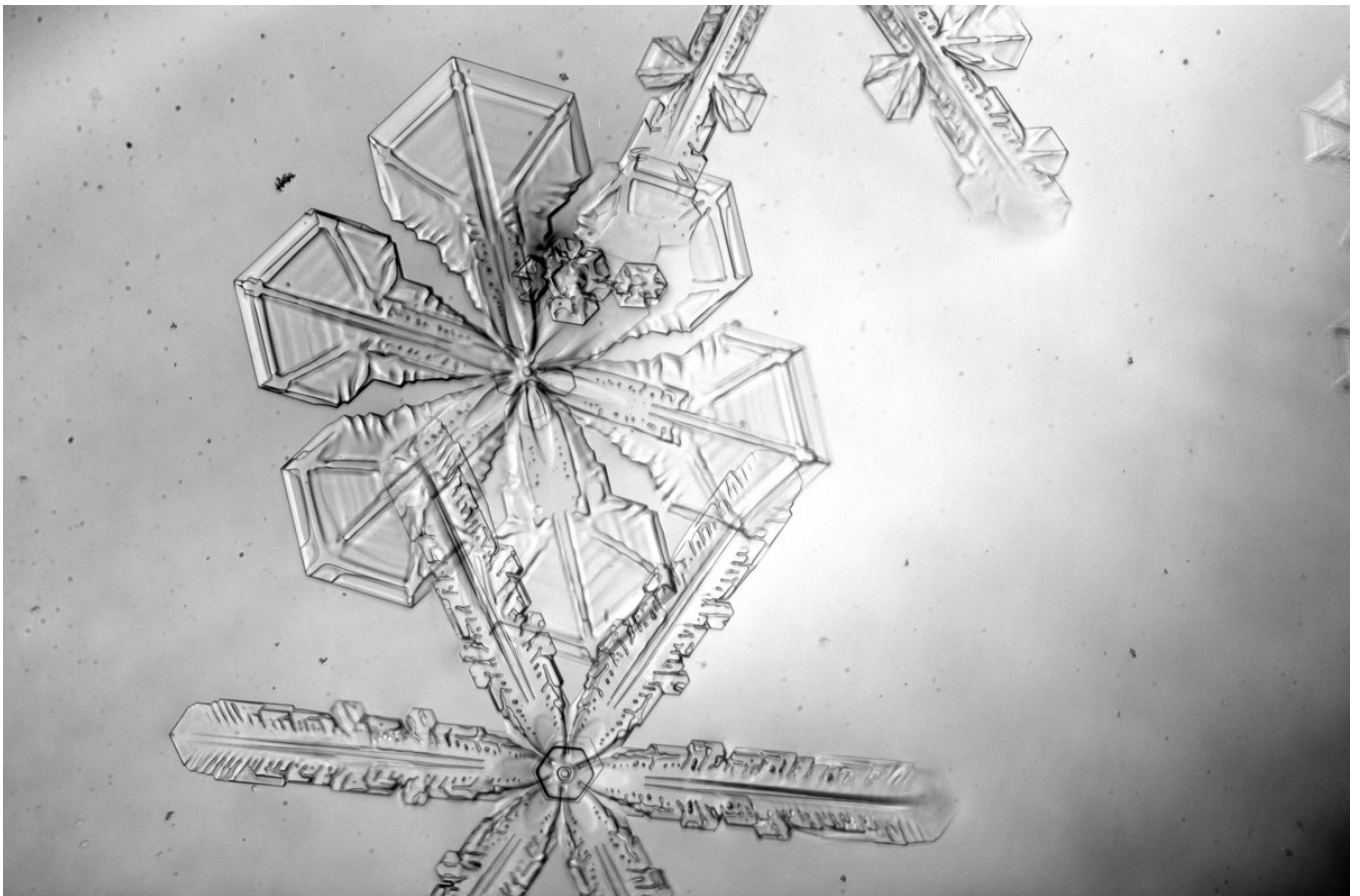
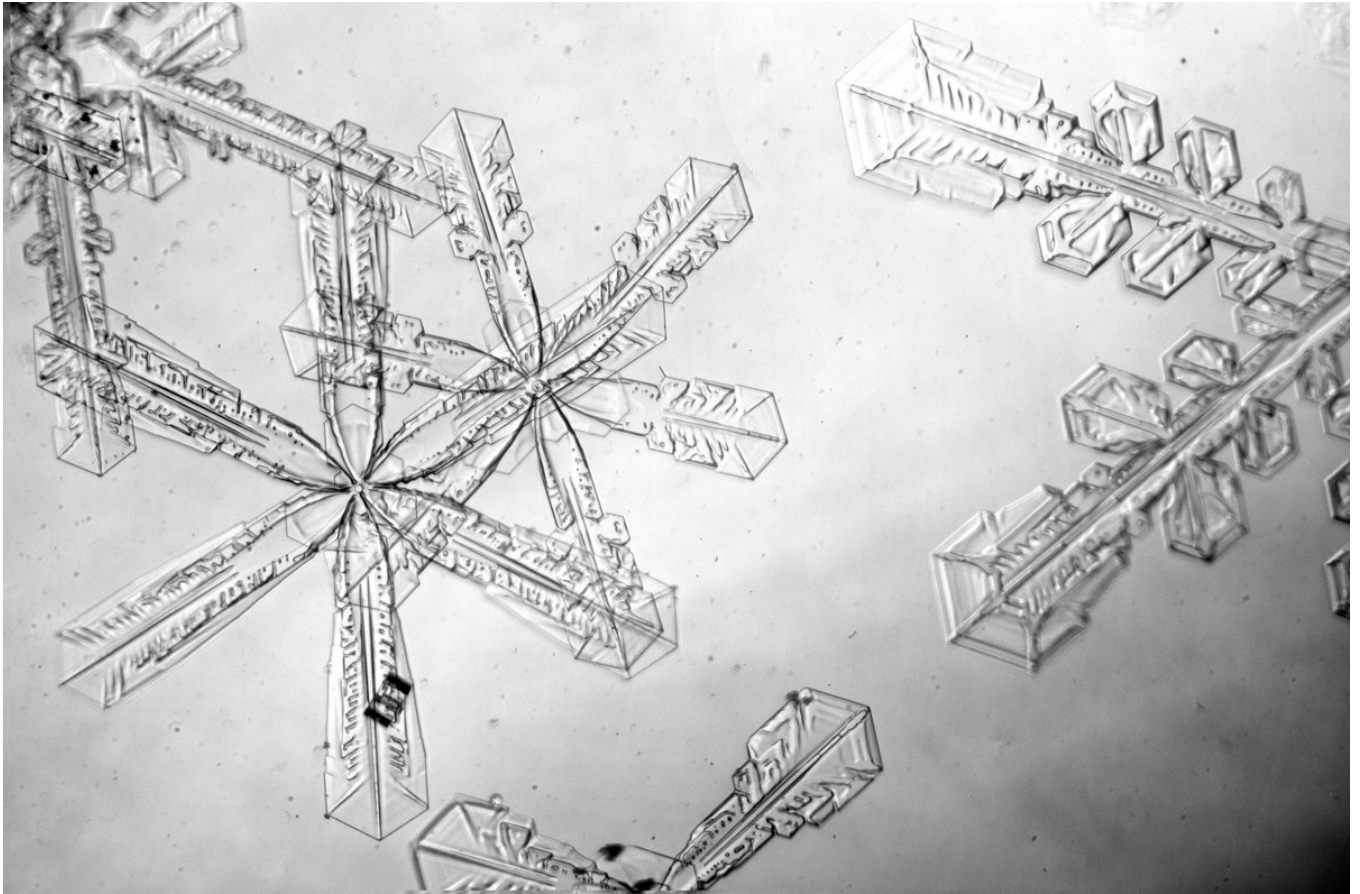
More crystals, 19 February Suosiola plume display.

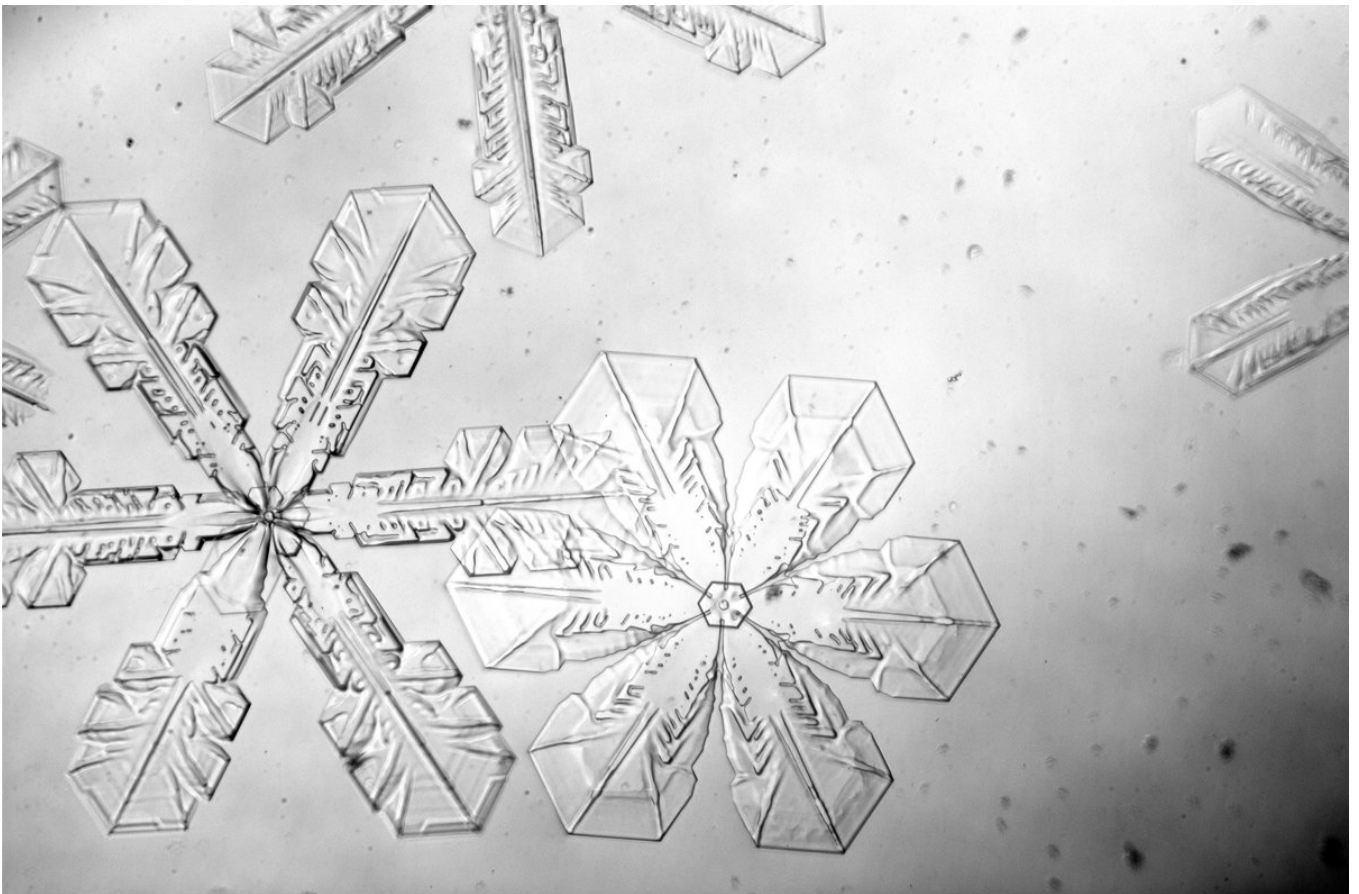
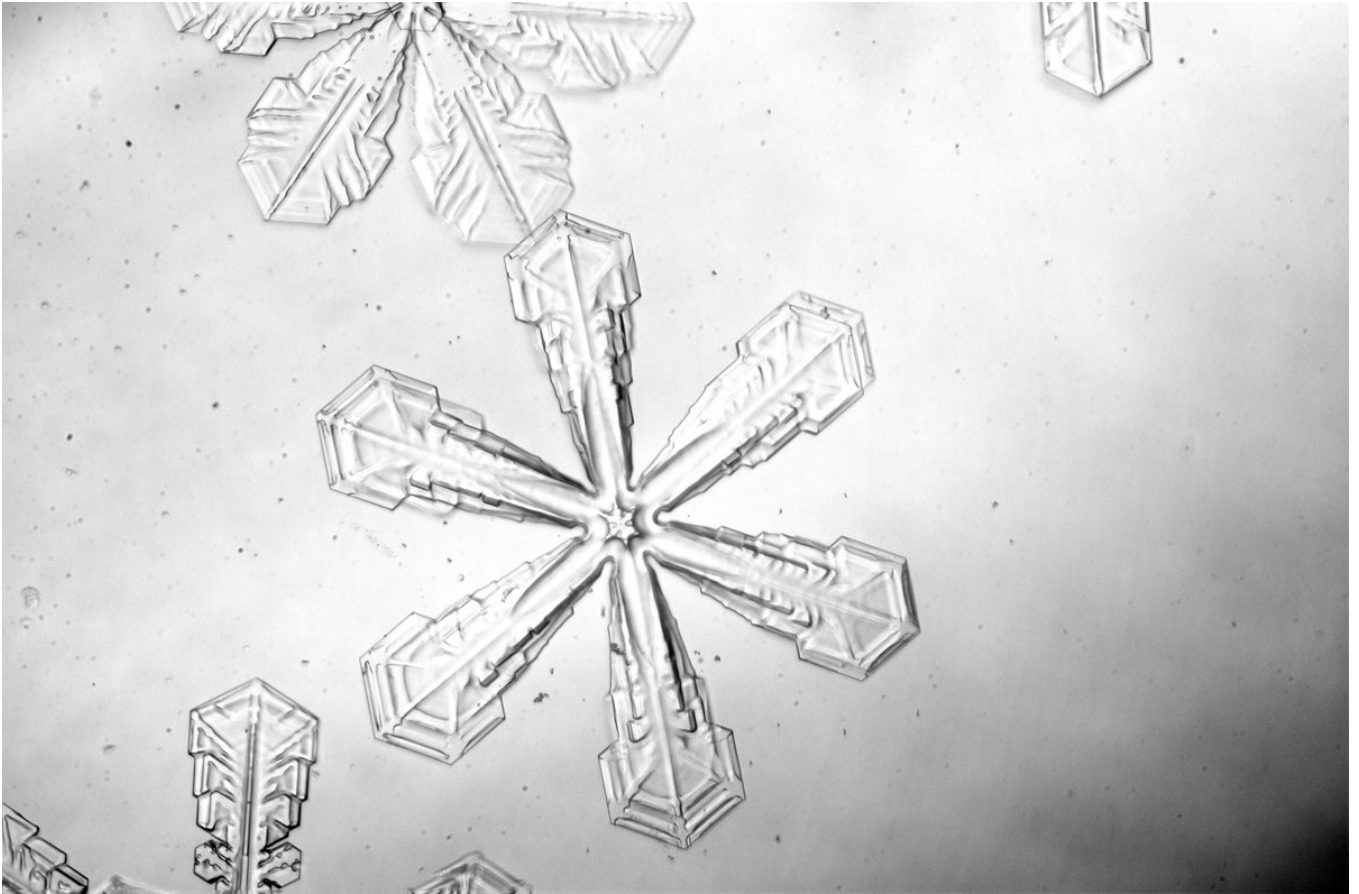




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More crystals, 19 February Suosiola plume display.





6/6 - 2024-02-23 15:41 UTC

Just needed to check for the possibility that the pillar threshold (19 February 2024 Suosiola plume

display) is an image processing artefact. First image shows a stack of some 10 successive frames in 2s interval as three versions: naturel, 30px, 100px usm (amount 500%). Second image has a selection of four single frames from that series in 4s interval to allow for the swarm to rotate its mass enough. Upper row naturel, lower 100px usm. Treatments in Photoshop.

To me this looks pretty conclusive – the threshold is real. But what it means? Maybe nothing. I remembered that in simulations pillar actually has a distinct boundary, as shown by the third image (HaloPoint, 0.01 h/d plates, 10 degs tilts, sun elev as in photos 12 degs). It is not nearly exact match, but snow stars are quite far removed from the idealized crystals of simulations. Their surface is uneven, internal raypaths are shadowed differently.

Anyway, looking at the images here, it looks to me the prospects are also the gap is real. And that is something I have hard time to see our simulations being able to produce any semblance of.



